

hw0324

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4.25 ## Data and notation

The variable `price` is interpreted as house price in thousands of dollars, and `sqft` is interpreted as hundreds of square feet. Therefore, a 2700-square-foot house has `sqft = 27`.

```
if (file.exists("collegetown.csv")) {
  data_path <- "collegetown.csv"
} else if (file.exists("/mnt/data/collegetown.csv")) {
  data_path <- "/mnt/data/collegetown.csv"
} else {
  stop("Please put collegetown.csv in the same folder as this Rmd file.")
}

ct <- read.csv(data_path)
names(ct) <- tolower(names(ct))

head(ct)
```

```
##   price  sqft age pool fireplace close twostory occupied
## 1  96.5 16.77 10    0          0      0          0          0
## 2 106.0 17.51 10    0          0      0          0          0
## 3 114.0 18.37  9    0          0      0          0          1
## 4 119.9 29.10  9    0          0      0          0          0
## 5 122.7 14.68  9    0          0      0          0          1
## 6 125.0 18.23  8    0          0      0          0          0
```

```
summary(ct[, c("price", "sqft")])
```

```
##      price      sqft
## Min.   : 50   Min.   :10.0
## 1st Qu.: 158   1st Qu.:20.6
## Median : 200   Median :25.5
## Mean   : 250   Mean   :27.3
## 3rd Qu.: 279   3rd Qu.:31.2
## Max.   :1370   Max.   :91.7
```

```
# Jarque--Bera test implemented directly to avoid extra package dependencies.
jb_test <- function(e) {
  n <- length(e)
  m <- mean(e)
  m2 <- mean((e - m)^2)
  m3 <- mean((e - m)^3)
  m4 <- mean((e - m)^4)
  skew <- m3 / (m2^(3/2))
  kurt <- m4 / (m2^2)
  jb <- n / 6 * (skew^2 + (kurt - 3)^2 / 4)
  pval <- 1 - pchisq(jb, df = 2)
  data.frame(skewness = skew, kurtosis = kurt, JB = jb, p.value = pval)
}

gen_r2 <- function(y, yhat_original_scale) {
  cor(y, yhat_original_scale)^2
}

fmt <- function(x, digits = 4) formatC(x, digits = digits, format = "f")
```

1 (a) Log-linear model

Estimate

$$\ln(PRICE_i) = \beta_1 + \beta_2 SQFT_i + e_i.$$

```
m_a <- lm(log(price) ~ sqft, data = ct)
summary(m_a)
```

```
##
## Call:
## lm(formula = log(price) ~ sqft, data = ct)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3253 -0.1920  0.0012  0.2168  0.8661
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)  4.39387    0.04329   101.5 <0.0000000000000002 ***
## sqft         0.03604    0.00149    24.3 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.34 on 498 degrees of freedom
## Multiple R-squared:  0.542, Adjusted R-squared:  0.541
## F-statistic: 589 on 1 and 498 DF, p-value: <0.0000000000000002
```

```

b1 <- coef(m_a)[["(Intercept)"]]
b2 <- coef(m_a)[["sqft"]]
price_bar <- mean(ct$price)
sqft_bar <- mean(ct$sqft)

# For log-linear model: dPRICE/dSQFT = beta2 * PRICE.
a_slope_mean <- b2 * price_bar
# Elasticity = beta2 * SQFT.
a_elast_mean <- b2 * sqft_bar
# Exact percentage change for one more unit of SQFT.
a_exact_pct <- 100 * (exp(b2) - 1)
a_approx_pct <- 100 * b2

ans_a <- data.frame(
  quantity = c("beta_1", "beta_2", "approx % change per 100 sq ft", "exact % change per 100 sq ft", "slope at sample means", "elasticity at sample means"),
  value = c(b1, b2, a_approx_pct, a_exact_pct, a_slope_mean, a_elast_mean)
)
knitr::kable(ans_a, digits = 4)

```

quantity	value
beta_1	4.3939
beta_2	0.0360
approx % change per 100 sq ft	3.6044
exact % change per 100 sq ft	3.6702
slope at sample means	9.0197
elasticity at sample means	0.9834

Answer. The estimated equation is

$$\ln(\widehat{PRICE}) = 4.3939 + 0.0360SQFT.$$

Since SQFT is measured in hundreds of square feet, one additional 100 square feet is associated with about 3.60% higher price, or 3.67% using the exact log interpretation. At the sample means, the slope is about 9.02 thousand dollars per 100 square feet, and the elasticity is about 0.983.

2 (b) Log-log model

Estimate

$$\ln(PRICE_i) = \alpha_1 + \alpha_2 \ln(SQFT_i) + e_i.$$

```

m_b <- lm(log(price) ~ log(sqft), data = ct)
summary(m_b)

```

```
##
## Call:
## lm(formula = log(price) ~ log(sqft), data = ct)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3686 -0.2085  0.0049  0.2320  1.2110
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)    2.0497     0.1580    13.0 <0.0000000000000002 ***
## log(sqft)      1.0248     0.0484    21.2 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.364 on 498 degrees of freedom
## Multiple R-squared:  0.474, Adjusted R-squared:  0.473
## F-statistic: 448 on 1 and 498 DF, p-value: <0.0000000000000002
```

```
a1 <- coef(m_b)[["(Intercept)"]]
a2 <- coef(m_b)[["log(sqft)"]]

# For log-log model: alpha2 is the elasticity.
b_elast <- a2
# Slope at means: dPRICE/dSQFT = alpha2 * PRICE/SQFT.
b_slope_mean <- a2 * price_bar / sqft_bar

ans_b <- data.frame(
  quantity = c("alpha_1", "alpha_2 / elasticity", "slope at sample means"),
  value = c(a1, a2, b_slope_mean)
)
knitr::kable(ans_b, digits = 4)
```

quantity	value
alpha_1	2.050
alpha_2 / elasticity	1.025
slope at sample means	9.400

Answer. The estimated equation is

$$\ln(\widehat{PRICE}) = 2.0497 + 1.0248 \ln(SQFT).$$

The coefficient 1.025 is an elasticity: a 1% increase in square footage is associated with about a 1.025% increase in price. At the sample means, the slope is about 9.40 thousand dollars per 100 square feet.

3 (c) Linear model and generalized R^2

Estimate the linear model

$$PRICE_i = \delta_1 + \delta_2 SQFT_i + e_i.$$

For the log models, I use the generalized R^2 defined as

$$R_g^2 = \left[\text{corr}(\text{PRICE}, \widehat{\text{PRICE}}) \right]^2,$$

where the log-model fitted values are transformed back to the original price scale.

```
m_c <- lm(price ~ sqft, data = ct)
summary(m_c)
```

```
##
## Call:
## lm(formula = price ~ sqft, data = ct)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -316.9  -58.9   -3.8   47.9  477.0
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)  -115.424     13.088   -8.82 <0.0000000000000002 ***
## sqft          13.403       0.449   29.84 <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 103 on 498 degrees of freedom
## Multiple R-squared:  0.641, Adjusted R-squared:  0.641
## F-statistic: 890 on 1 and 498 DF, p-value: <0.0000000000000002
```

```
d1 <- coef(m_c)[["(Intercept)"]]
d2 <- coef(m_c)[["sqft"]]

c_slope <- d2
c_elast_mean <- d2 * sqft_bar / price_bar

r2_linear <- summary(m_c)$r.squared
r2_gen_a <- gen_r2(ct$price, exp(fitted(m_a)))
r2_gen_b <- gen_r2(ct$price, exp(fitted(m_b)))

r2_table <- data.frame(
  model = c("Linear: price ~ sqft", "Log-linear: log(price) ~ sqft", "Log-log: log(pr
ice) ~ log(sqft)"),
  R2_type = c("ordinary R2", "generalized R2", "generalized R2"),
  R2 = c(r2_linear, r2_gen_a, r2_gen_b)
)
knitr::kable(r2_table, digits = 4)
```

model	R2_type	R2
Linear: price ~ sqft	ordinary R2	0.6413
Log-linear: log(price) ~ sqft	generalized R2	0.6622
Log-log: log(price) ~ log(sqft)	generalized R2	0.6445

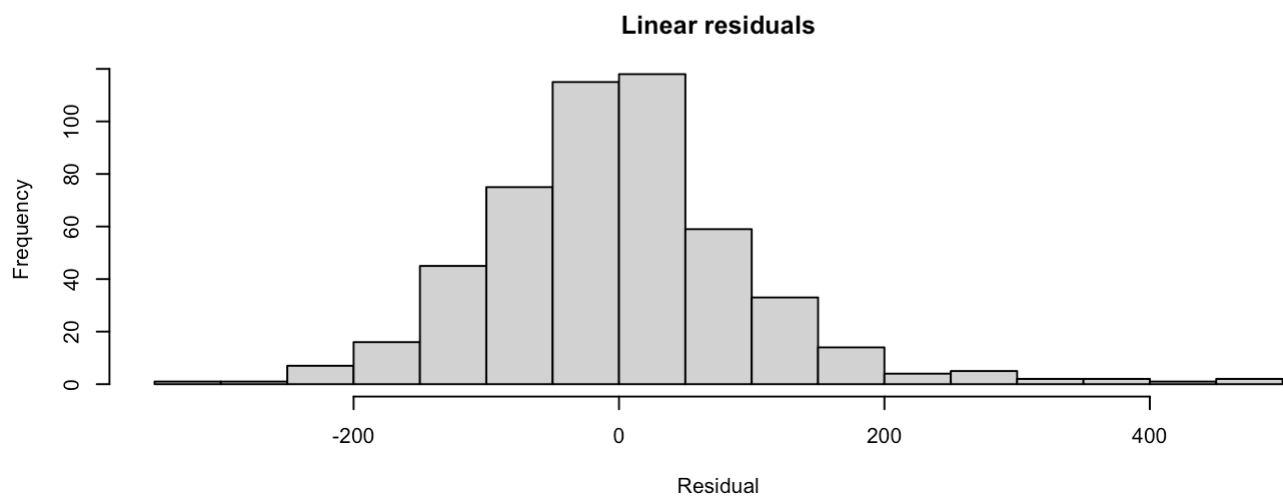
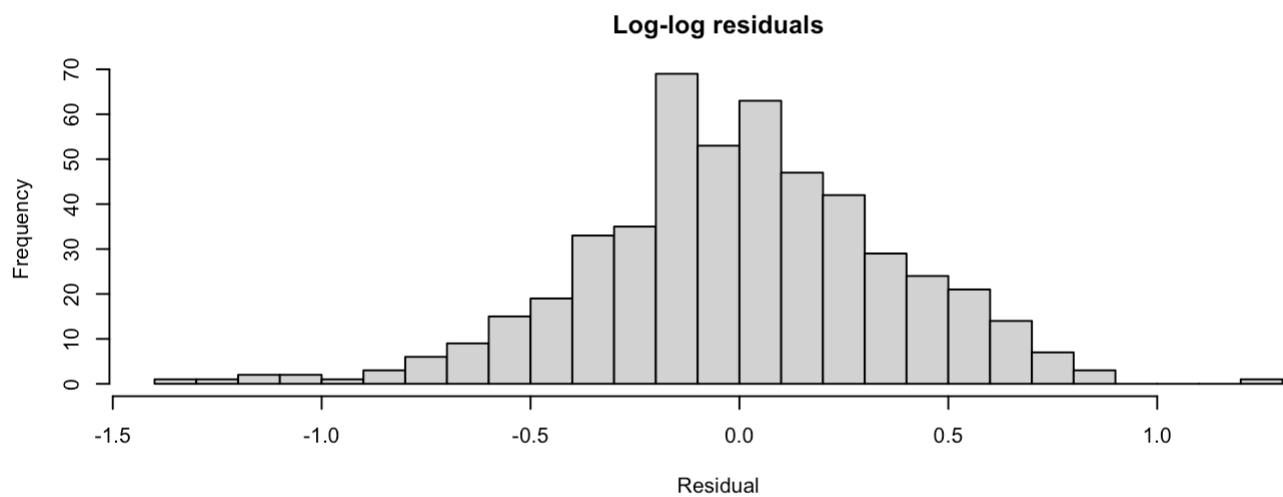
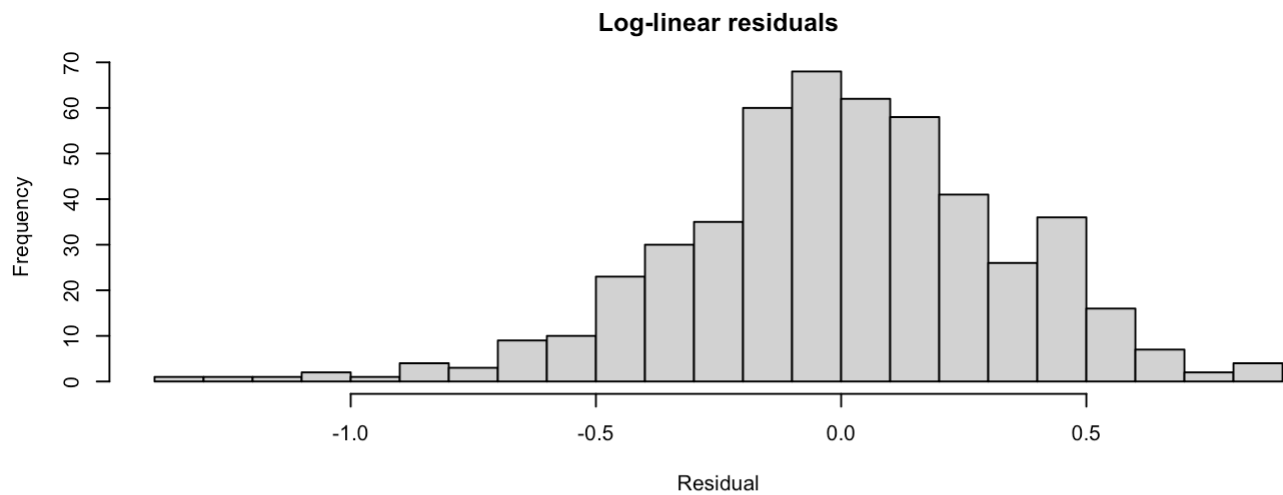
```
linear_effect_table <- data.frame(
  quantity = c("delta_1", "delta_2 / slope", "elasticity at sample means"),
  value = c(d1, d2, c_elast_mean)
)
knitr::kable(linear_effect_table, digits = 4)
```

quantity	value
delta_1	-115.424
delta_2 / slope	13.403
elasticity at sample means	1.461

Answer. The linear model gives an ordinary R^2 of 0.6413. The generalized R^2 values are 0.6622 for the log-linear model and 0.6445 for the log-log model. Based only on this criterion, the log-linear model fits slightly better on the original price scale, although the differences are not large.

4 (d) Histograms and Jarque–Bera tests

```
par(mfrow = c(3, 1), mar = c(4, 4, 3, 1))
hist(resid(m_a), breaks = 25, main = "Log-linear residuals", xlab = "Residual")
hist(resid(m_b), breaks = 25, main = "Log-log residuals", xlab = "Residual")
hist(resid(m_c), breaks = 25, main = "Linear residuals", xlab = "Residual")
```



```
par(mfrow = c(1, 1))

jb_table <- rbind(
  `Log-linear` = jb_test(resid(m_a)),
  `Log-log` = jb_test(resid(m_b)),
  `Linear` = jb_test(resid(m_c))
)
knitr::kable(jb_table, digits = 4)
```

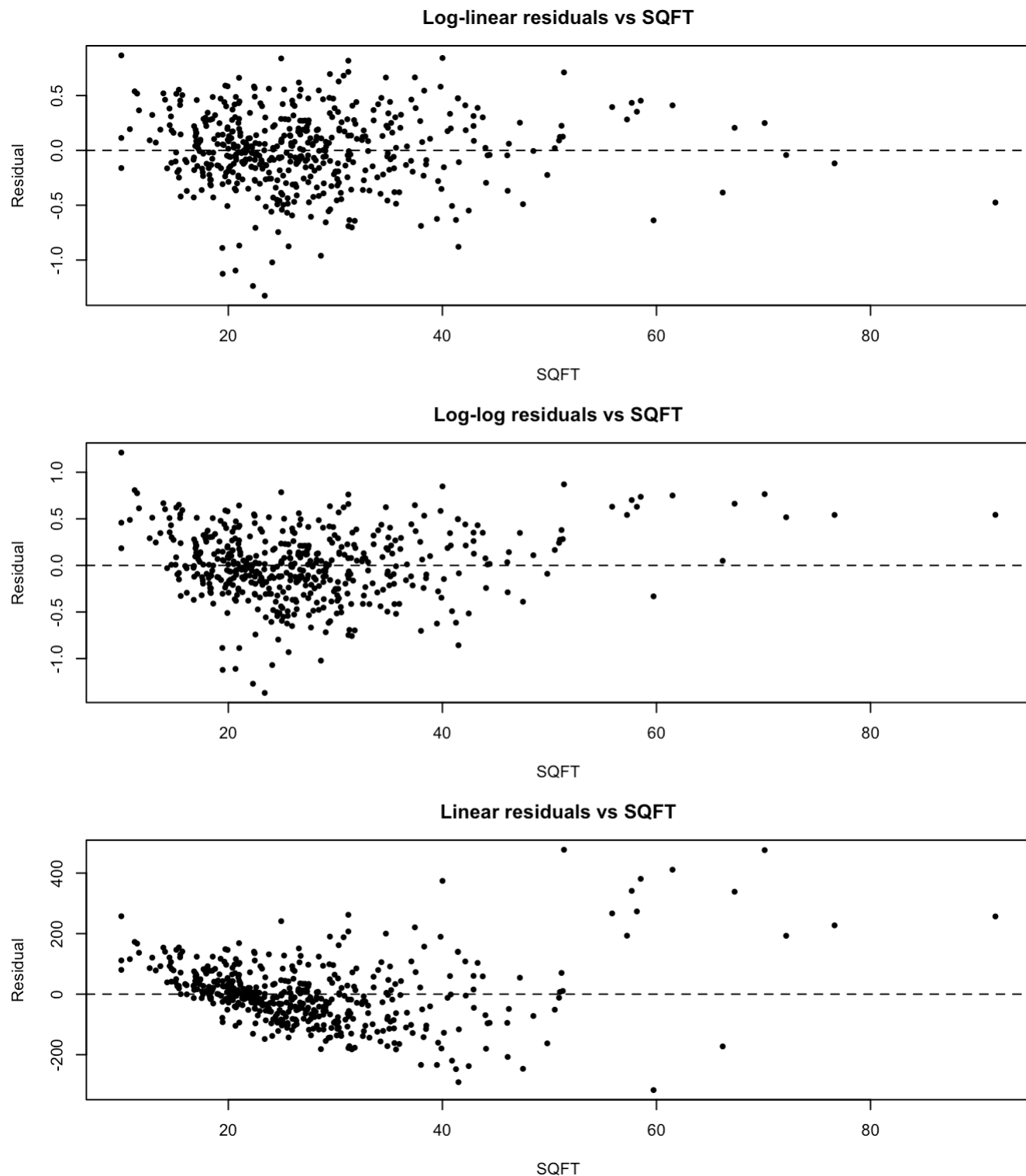
	skewness	kurtosis	JB	p.value
Log-linear	-0.4073	3.785	26.68	0.0000

	skewness	kurtosis	JB	p.value
Log-log	-0.2476	3.663	14.28	0.0008
Linear	0.8049	5.832	221.11	0.0000

Answer. The Jarque–Bera p-values are small for all three models, so the residuals are not fully compatible with normality under a formal test. The linear model has the strongest non-normality, while the log-log model looks the closest to normal among the three.

5 (e) Residuals against SQFT

```
par(mfrow = c(3, 1), mar = c(4, 4, 3, 1))
plot(ct$sqft, resid(m_a), pch = 19, cex = 0.6,
     main = "Log-linear residuals vs SQFT", xlab = "SQFT", ylab = "Residual")
abline(h = 0, lty = 2)
plot(ct$sqft, resid(m_b), pch = 19, cex = 0.6,
     main = "Log-log residuals vs SQFT", xlab = "SQFT", ylab = "Residual")
abline(h = 0, lty = 2)
plot(ct$sqft, resid(m_c), pch = 19, cex = 0.6,
     main = "Linear residuals vs SQFT", xlab = "SQFT", ylab = "Residual")
abline(h = 0, lty = 2)
```

```
par(mfrow = c(1, 1))
```

Answer. The linear model shows a clear pattern and larger dispersion for large houses, suggesting heteroskedasticity and possible functional-form problems. The log models reduce this problem. The log-linear model has a relatively flatter residual pattern, while the log-log model shows a mild curved pattern.

6 (f) Prediction for a 2700-square-foot house

Because `sqft` is measured in hundreds of square feet, use `sqft = 27`.

```
new_house <- data.frame(sqft = 27)

pred_a_log <- predict(m_a, newdata = new_house)
pred_b_log <- predict(m_b, newdata = new_house)
pred_c <- predict(m_c, newdata = new_house)

pred_table <- data.frame(
  model = c("Log-linear", "Log-log", "Linear"),
  predicted_price_thousand = c(exp(pred_a_log), exp(pred_b_log), pred_c)
)
knitr::kable(pred_table, digits = 3)
```

model	predicted_price_thousand
Log-linear	214.2
Log-log	227.5
Linear	246.5

Answer. The predicted prices for a 2700-square-foot house are shown in the table above. The predictions are in thousands of dollars.

7 (g) 95% prediction intervals for a 2700-square-foot house

```
pi_a_log <- predict(m_a, newdata = new_house, interval = "prediction", level = 0.95)
pi_b_log <- predict(m_b, newdata = new_house, interval = "prediction", level = 0.95)
pi_c <- predict(m_c, newdata = new_house, interval = "prediction", level = 0.95)

pi_table <- data.frame(
  model = c("Log-linear", "Log-log", "Linear"),
  fit = c(exp(pi_a_log[, "fit"]), exp(pi_b_log[, "fit"]), pi_c[, "fit"]),
  lower_95_PI = c(exp(pi_a_log[, "lwr"]), exp(pi_b_log[, "lwr"]), pi_c[, "lwr"]),
  upper_95_PI = c(exp(pi_a_log[, "upr"]), exp(pi_b_log[, "upr"]), pi_c[, "upr"])
)
knitr::kable(pi_table, digits = 3)
```

model	fit	lower_95_PI	upper_95_PI
Log-linear	214.2	109.77	418.1
Log-log	227.5	111.14	465.8
Linear	246.5	44.28	448.6

Answer. The prediction intervals are in thousands of dollars. For the log models, the intervals are first computed on the log-price scale and then exponentiated back to the price scale.

8 (h) Choice of functional form

Answer. I would use the log-linear model as the main specification. It has the highest generalized R^2 on the original price scale, and its residual plot against SQFT shows less systematic curvature than the log-log model and much less pattern than the linear model. However, the Jarque–Bera tests reject normality for all models, so any inference based on exact residual normality should be interpreted with caution.